

Ganit

data speaks

Walmart – Data Warehouse & Tableau BI Platform

Case Study

Unified Sales, Inventory & Category Analytics on AWS — Enabling Real-Time Cross-Functional Decisions Across Stores, Channels & Supply Chain

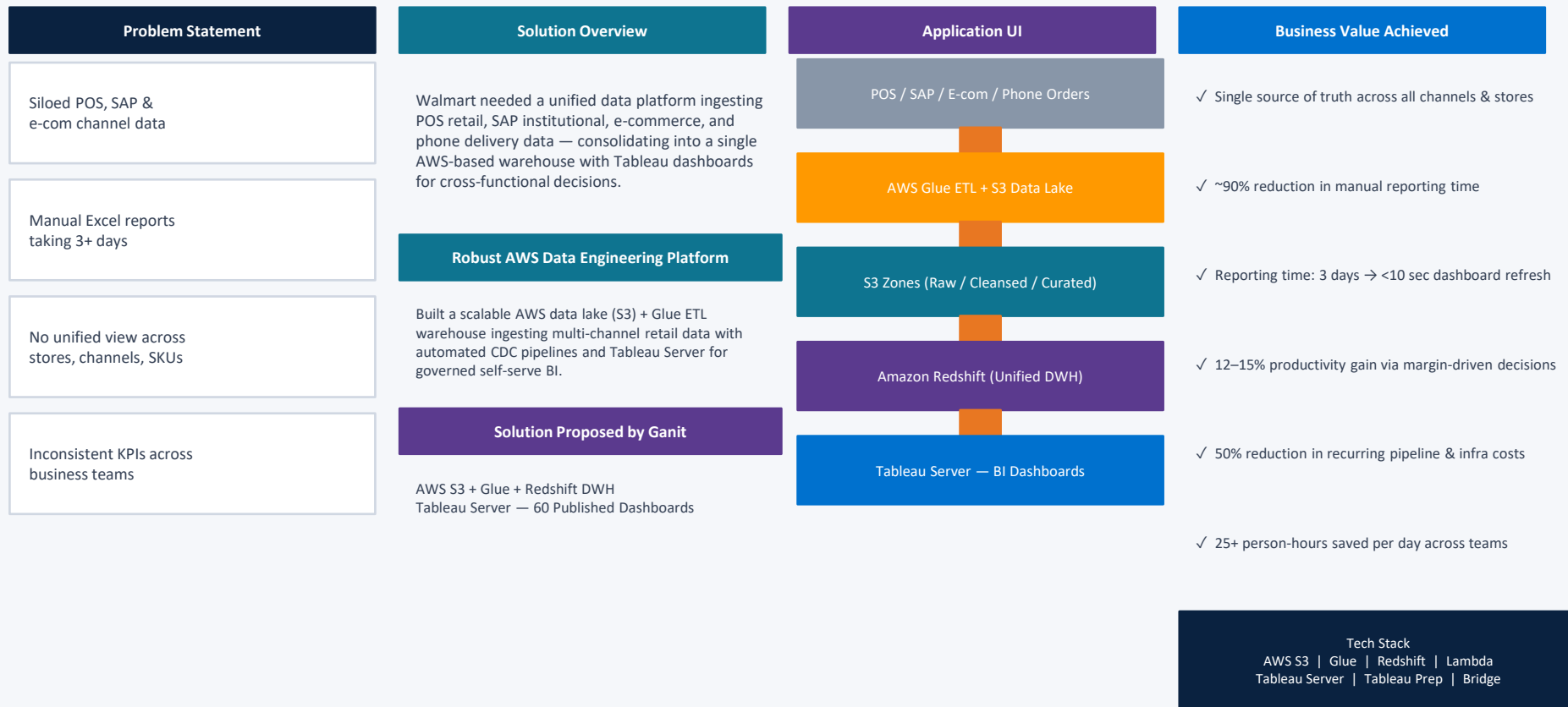
Retail | Omnichannel

DE | AWS

BI | Tableau

10,500+ Stores | Global

Building a Unified Retail Data Warehouse on AWS and Self-Serve Tableau BI Platform for Walmart — Enabling Real-Time Sales, Inventory & Category Analytics Across Stores, Channels & Supply Chain



Ganit unified Walmart's retail, e-com & institutional sales data on AWS, delivering 60 Tableau dashboards that transformed manual reporting into real-time cross-functional intelligence across stores, channels & supply chain

Outcome

Delivered a cloud-native AWS data warehouse ingesting POS retail, SAP institutional, e-commerce portal, and phone delivery data through automated AWS Glue ETL pipelines into a unified Amazon Redshift DWH — governed with CDC pipelines, DQ checks, and custom calendar logic — surfaced via Tableau Server for 18 business teams with 60 dashboards covering 1,470 KPIs.

Operational Efficiency

Eliminated 3-day manual Excel consolidation by automating multi-channel ingestion via AWS Glue — reducing reporting latency from 3 days to sub-10-second Tableau dashboard refresh across all stores.

Cost Optimisation

S3 file partitioning best practices reduced PUT/GET/LIST requests significantly — achieving 50% reduction in recurring pipeline costs. Single published Tableau extracts power multiple dashboards, removing redundant processing.

Scalability & Reliability

Built a three-zone S3 data lake (Raw / Cleansed / Curated) handling 800+ SFTP files per month, 5 API connections, 6 source systems, and 28 CDC pipelines — with automated DQ checks pre-ingestion.

Business Growth

60 Tableau dashboards covering Sales, Category, SCM, PnL, Marketing and Customer Analytics gave 18 business teams a single source of truth — improving target achievement compliance by 25% and productivity by 12–15%.

Siloed POS, SAP & e-commerce data, inconsistent KPIs across teams, and manual Excel-based reporting blocked unified retail analytics across Walmart's store network and supply chain

Problem Statement

Walmart operates one of the world's largest retail networks — spanning physical stores, e-commerce, institutional sales, and phone delivery — running on a complex mix of POS systems, SAP ERP, and third-party e-com portals.

Sales data was fragmented across independent systems with each channel reporting at different granularities. Every team maintained separate Excel files for the same KPIs, resulting in conflicting numbers, 3-day reporting delays, and no ability to analyse performance across Product, Location, and Channel hierarchies simultaneously. Manual processes consumed 25+ person-hours per day, and there was no mechanism for consistent target tracking, custom time-period comparisons, or automated data quality assurance.

Challenge 1

POS retail, SAP institutional, e-commerce & phone delivery data siloed — no unified ingestion or reconciliation pipeline

Challenge 2

Each business team had different numbers for the same KPI — no single source of truth across Sales, NOB, ABV, GM%

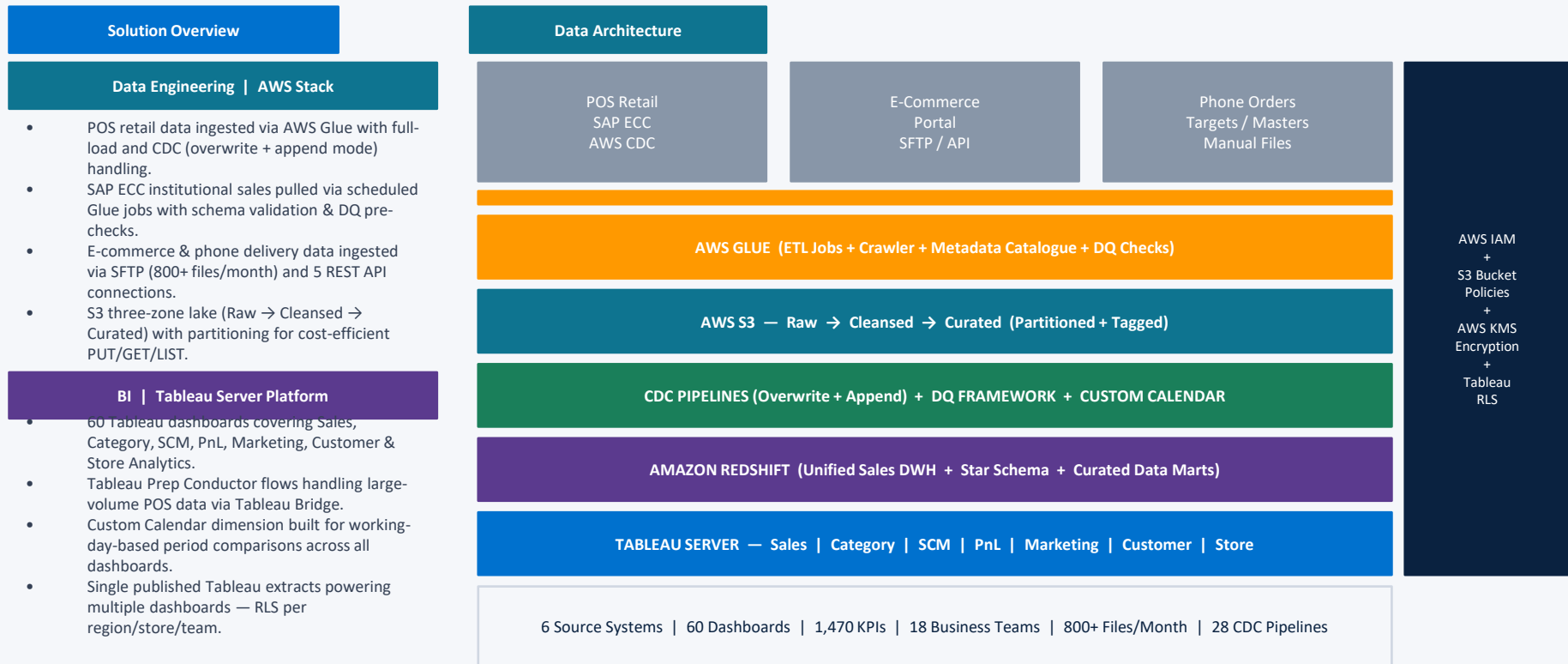
Challenge 3

Custom time-period comparison (working day-based, not date-based) not available — blocking accurate period-over-period analysis

Challenge 4

NOB non-linear aggregation logic differed across Store and Product hierarchies — causing incorrect reporting at multiple levels

Ganit built a multi-source AWS data warehouse for Walmart with automated Glue ETL pipelines, CDC ingestion, and a 60-dashboard Tableau Server platform for real-time retail analytics



Volumetrics for Walmart's Engagement

Business Intelligence | Tableau

5 Business Functions

18 Business Teams

60 Reports

45 Curated Tables

60 Dashboards

1,470 KPIs

Custom Calendar

Store Analytics

RLS + Row Security

Data Engineering | AWS

6 Source Systems

5 API Connections

800+ Files/Month (SFTP)

100+ Bronze Tables

75 Silver Tables

50 Gold Tables

35 Daily Pipelines

28 CDC Pipelines

DQ & Custom Checks

Combined Engagement Impact

~90%

Reduction in
Manual Reporting3 Days
→ <10 secDashboard
Refresh Speed

25+

Person-Hours
Saved Daily

50%

Cost Reduction
(Pipelines + Infra)

12–15%

Productivity
Improvement